

Map Setup and Prop Installation

Preface:

This document pertains to the autonomous driving capabilities of our company's vehicle. Before engaging the autonomous driving feature, please verify that the hardware components (such as the depth camera and car chassis motors) installed in the vehicle meet the specified requirements and are operating correctly. Additionally, ensure that the battery is adequately charged.

1. Props Setup & Notices

Prior to commencing autonomous driving, please follow the steps below to set up the map.

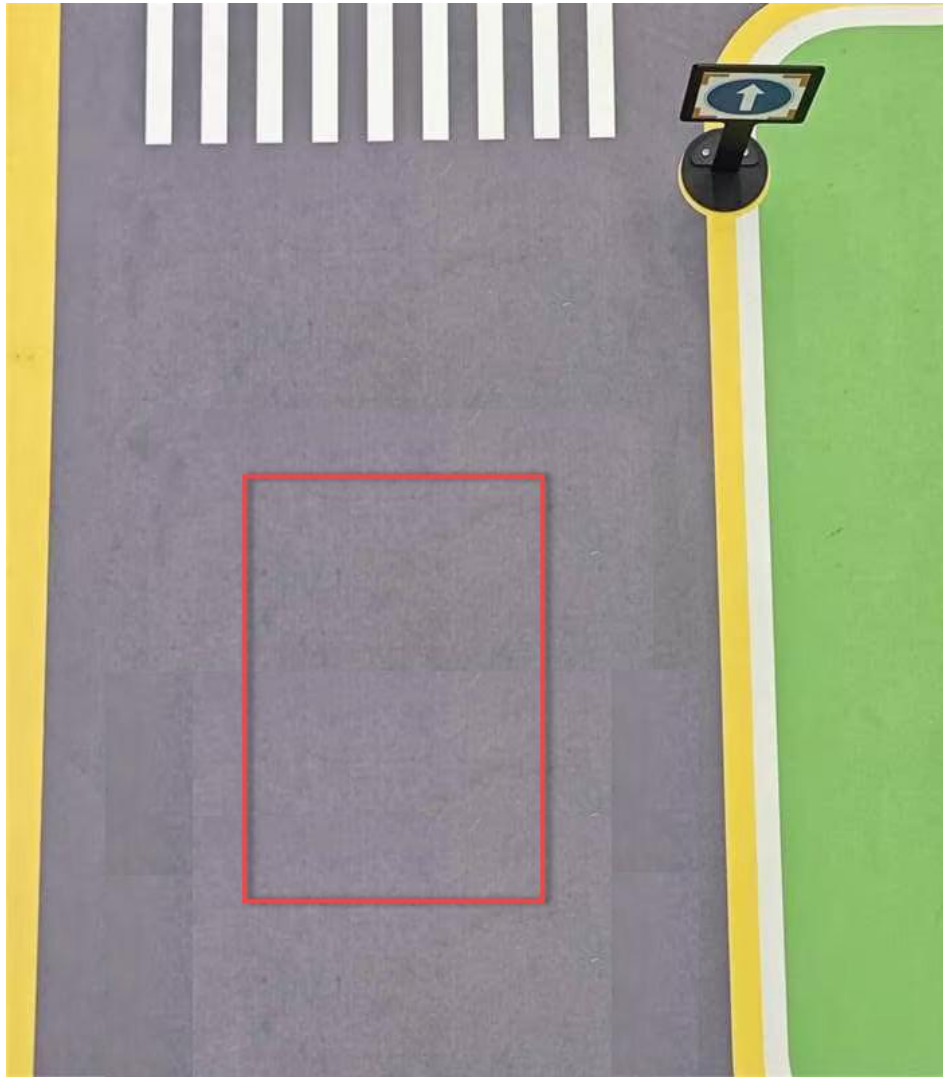
1.1 Setup Instructions

① Map Setup

To begin, ensure that the site is situated on a flat and spacious ground, and verify that there is ample lighting surrounding the area, as illustrated in the image below:



Place the robot at the start point as pictured.

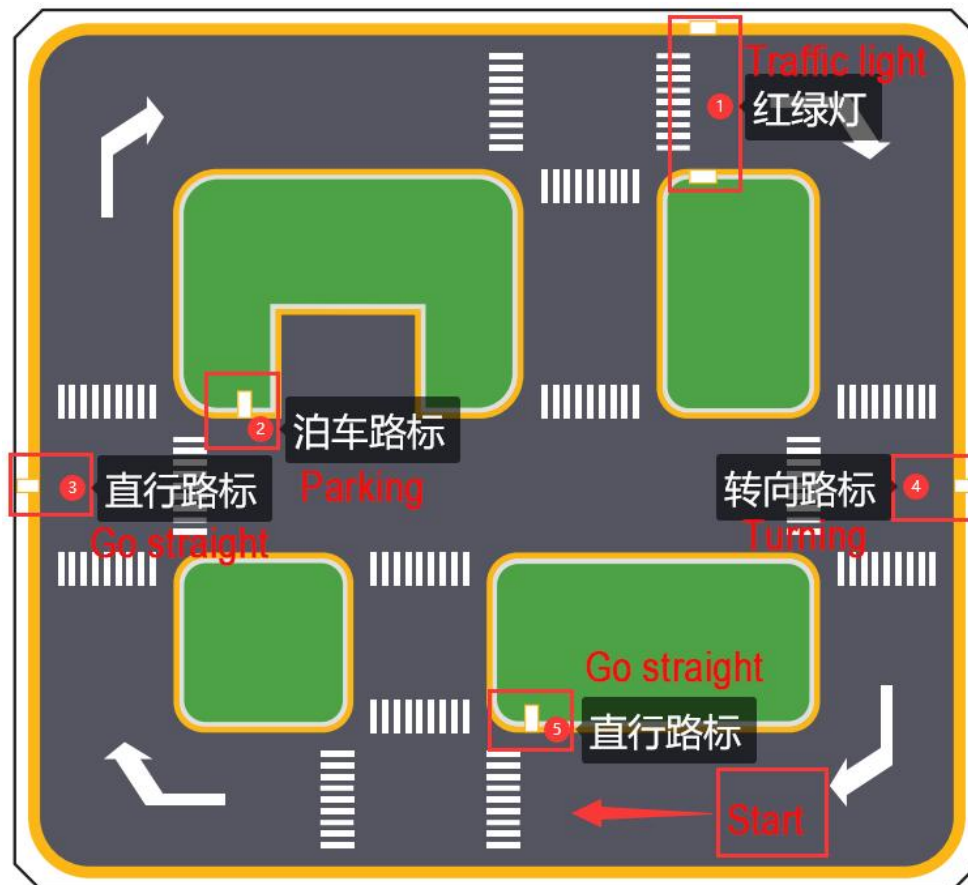


As the robot advances, it will navigate along the yellow line bordering the map, and it can dynamically adjust its posture based on the lane lines in real-time.

② Road Signs Setup

In the game of autonomous driving, a total of 4 waypoints need to be placed: 2 for straight driving, 1 for right turn, and 1 for parking. Please refer to the following image for the specific placement positions.

Note: Ensure accurate placement!!



Go straight: instruct the robot car to go forward.





Turn right: Instruct the robot car to turn right.



Park: Command the robot car to park.



The purpose of the road signs is to guide the car during its journey. Once the car recognizes a road sign, it will perform the appropriate actions. Additionally, the car will slow down as it approaches a pedestrian crosswalk.

③ Traffic Light Setup

In the game of the autonomous driving function, only one traffic light needs to be placed. The exact placement location is illustrated in the figure below:



Traffic light props are utilized to replicate real-world traffic lights on roads.

When the car detects the traffic light, it will adhere to the “stop on red light, go on green light” rule.

Once all venue props are arranged, refer to the following picture for setting up the venue environment:



1.2 Notice:

1. Ensure that the site has adequate lighting, preferably natural ambient light. Avoid strong direct light sources and colored lights, as they may hinder overall recognition. Pay attention to this aspect when setting up the site.
2. While laying out and using the equipment, ensure proper care and protection of the props on the site. In case of incomplete maps or signs, or if the traffic light is damaged and cannot function properly, promptly contact our after-sales personnel to purchase replacements to prevent any disruptions to the recognition process.